

Experiment 2.3:

1. Greatest Among Three Numbers

Program

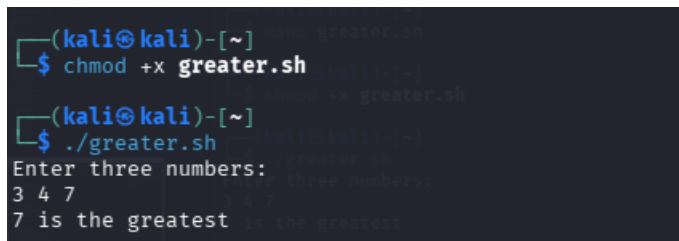
```
#!/bin/bash

echo "ENTER THREE NUMBERS"

read a b c

if [ $a -gt $b ] && [ $a -gt $c ]
then
echo "$a is greater"
elif [ $b -gt $c ]
then
echo "$b is greater"
else
echo "$c is greater"
fi
```

Output:



```
(kali㉿kali)-[~]
└─$ chmod +x greater.sh
(kali㉿kali)-[~]
└─$ ./greater.sh
Enter three numbers:
3 4 7
7 is the greatest
```

2. Factorial of a Given Number

Program

```
#!/bin/bash

echo "ENTER THE NUMBER:"

read n

fact=1

while [ $n -gt 1 ]
do
fact=$((fact * n))
n=$((n - 1))
```

done

echo "FACTORIAL OF THE GIVEN NUMBER IS \$fact"

Output:

```
(kali㉿kali)-[~]
└─$ chmod +x factorial.sh greater.sh

(kali㉿kali)-[~]
└─$ ./factorial.sh
ENTER THE NUMBER:
4
FACTORIAL OF THE GIVEN NUMBER IS 24
```

3: Sum of Odd Numbers up to N

Program

```
#!/bin/bash
```

```
echo "ENTER THE RANGE:"
```

```
read n
```

```
x=1
```

```
sum=0
```

```
while [ $x -le $n ]
```

```
do
```

```
sum=$((sum + x))
```

```
x=$((x + 2))
```

```
done
```

```
echo "SUM = $sum"
```

Output:

```
(kali㉿kali)-[~]
└─$ chmod +x oddsum.sh greater.sh

(kali㉿kali)-[~]
└─$ ./oddsum.sh
ENTER THE RANGE:
5
SUM = 9
```

4: Generation of Fibonacci Numbers

Program

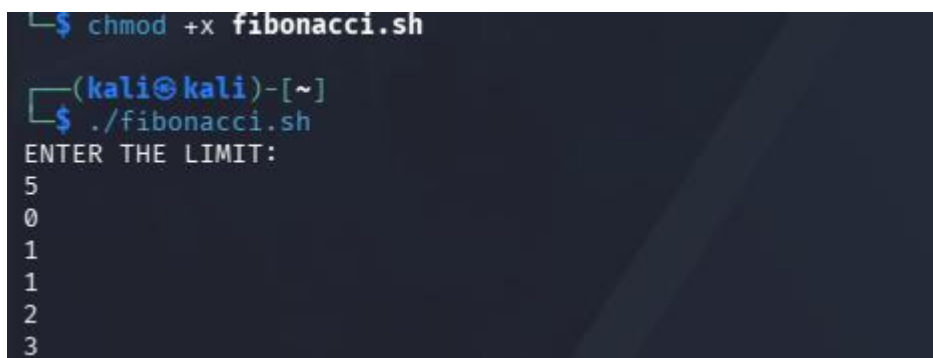
```
#!/bin/bash
```

```
echo "ENTER THE LIMIT:"
```

```
read n
```

```
p=-1
q=1
i=1
while [ $i -le $n ]
do
r=$((p + q))
p=$q
q=$r
echo "$r"
i=$((i + 1))
done
```

Output:

A terminal window with a dark background. The prompt is a blue dollar sign. The first command is 'chmod +x fibonacci.sh'. The second command is './fibonacci.sh'. The output is 'ENTER THE LIMIT:'. Below this, the numbers 5, 0, 1, 1, 2, 3 are printed on separate lines, representing the Fibonacci sequence up to the 5th term.

```
$ chmod +x fibonacci.sh
(kali@kali)-[~]
$ ./fibonacci.sh
ENTER THE LIMIT:
5
0
1
1
2
3
```

5: Arithmetic Calculator

Program

```
#!/bin/bash

echo "ENTER THE VALUE OF A:"
read a

echo "ENTER THE VALUE OF B:"
read b

echo "ENTER THE OPTION TO PERFORM"
echo "1. ADDITION"
echo "2. SUBTRACTION"
echo "3. MULTIPLICATION"
echo "4. DIVISION"
read op
```

```

case "$op" in
1) echo "Result = $((a + b))" ;;
2) echo "Result = $((a - b))" ;;
3) echo "Result = $((a * b))" ;;
4) echo "Result = $((a / b))" ;;
*) echo "Invalid Option" ;;

Esac

```

Output:

```

(kali㉿kali)-[~]
└─$ chmod +x calculator.sh
└─$ ./calculator.sh
./calculator.sh: 1: Program: not found
ENTER THE VALUE OF A:
6
ENTER THE VALUE OF B:
7
ENTER THE OPTION TO PERFORM:
1. ADDITION
2. SUBTRACTION
3. MULTIPLICATION
4. DIVISION
1
Result = 13

```

6: Largest Digit of a Number

Program

```

#!/bin/bash

echo "ENTER THE NUMBER"

read a

max=0

while [ $a -gt 0 ]

do

r=$((a % 10))

if [ $r -gt $max ]

then

max=$r

fi

a=$((a / 10))

done

```

```
echo "THE LARGEST DIGIT OF THE NUMBER: $max"
```

Output:

```
(kali㉿kali)-[~]
└─$ chmod +x largestdigit.sh
└─$ ./largestdigit.sh
./largestdigit.sh: 1: Program: not found
ENTER THE NUMBER
58329
THE LARGEST DIGIT OF THE NUMBER: 9
```

7: Palindrome String Check

Program

```
#!/bin/bash
```

```
echo "ENTER THE STRING TO CHECK PALINDROME"
```

```
read str
```

```
len=$(echo -n "$str" | wc -c)
```

```
i=1
```

```
j=$((len / 2))
```

```
while [ $i -le $j ]
```

```
do
```

```
k=$(echo "$str" | cut -c $i)
```

```
l=$(echo "$str" | cut -c $len)
```

```
if [ "$k" != "$l" ]
```

```
then
```

```
echo "$str is not a palindrome"
```

```
exit
```

```
fi
```

```
i=$((i + 1))
```

```
len=$((len - 1))
```

```
done
```

```
echo "$str is a palindrome"
```

Output:

```
(kali㉿kali)-[~]
$ chmod +x palindrome.sh
./palindrome.sh
./palindrome.sh: 1: Program: not found
ENTER THE STRING TO CHECK PALINDROME
madam
madam is a palindrome
```

8: Reverse of a Given Number

Program

```
#!/bin/bash

echo "ENTER THE NUMBER"

read n

rnum=0

while [ $n -ne 0 ]

do

remainder=$((n % 10))

rnum=$((rnum * 10 + remainder))

n=$((n / 10))

done

echo "REVERSE OF THE NUMBER IS $rnum"
```

Output:

```
(kali㉿kali)-[~]
$ nano reverse.sh
$ ./oddsum.sh
ENTER THE RANGE:
54321
SUM = 737719921
```

